There is no such tendency shown by Cl^- and therefore $[H^+] > [OH^-]$; the solution is acidic and the pH is less than 7.

As discussed in the salt hydrolysis of strong base and weak acid. In this case also, we can establish a relationship between the K_h and K_b as

$$K_h . K_b = K_w$$

Let us calculate the K_h value in terms of degree of hydrolysis (h) and the concentration of salt

$$K_h = h^2 C \qquad \text{and } [H^+] = \sqrt{K_h . C}$$

$$[H^+] = \sqrt{\frac{K_w}{K_b} . C}$$

$$pH = - \log [H^+]$$

$$= - \log \left(\frac{K_w . C}{K_b} \right)^{1/2}$$

$$= - \frac{1}{2} \log K_w - \frac{1}{2} \log C + \frac{1}{2} \log K_b$$

$$pH = 7 - \frac{1}{2} pK_b - \frac{1}{2} \log C.$$

8.8.4 Hydrolysis of Salt of weak acid and weak base (Anionic & Cationic Hydrolysis).

Let us consider the hydrolysis of ammonium acetate.

$$CH_3COONH_4(aq) \rightarrow CH_3COO^-(aq) + NH_4^+(aq)$$

In this case, both the cation (NH_4^+) and anion (CH_3COO^-) have the tendency to react with water

$$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$$

$$NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$$

The nature of the solution depends on the strength of acid (or) base i.e, if $K_a > K_b$; then the solution is acidic and pH < 7, if $K_a < K_b$; then the solution is basic and pH > 7, if $K_a = K_b$; then the solution is neutral.

The relation between the dissociation constant (K_a, K_b) and the hydrolysis constant is given by the following expression.

$$K_a . K_b . K_h = K_w$$

pH of the solution

pH of the solution can be calculated using the following expression,

$$pH = 7 + \frac{1}{2} pK_a - \frac{1}{2} pK_b .$$

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